

Grade 3 lessons

Times tables, fractions and area

IN THIS BOOK

1. Multiplication
2. Division
3. Fractions
4. Area and perimeter
5. Rounding

HOW TO USE IT

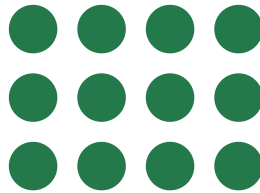
Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 3 Kahoot and Blooket sets, and one stop on the Grade 3 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

Multiplication

THE BIG IDEA

Multiplying is a fast way to add equal groups. 4×3 means 4 groups of 3.



3 rows with 4 in each row: $3 \times 4 = 12$.

HOW TO DO IT

1. Read $a \times b$ as "a groups of b".
2. Skip count by b, a times. Or picture rows and columns.
3. You can swap the order. 4×3 gives the same answer as 3×4 .
4. Learn the times tables so the answers come quickly.

WORKED EXAMPLES

$4 \times 3 = ?$

Count by 3s four times: 3, 6, 9, 12. So $4 \times 3 = 12$.

$6 \times \square = 42$

Count by 6s until you reach 42: 6, 12, 18, 24, 30, 36, 42. That is 7 counts. The missing number is 7.

WATCH OUT

Multiplying is not adding. 4×3 is 12, but $4 + 3$ is only 7.

WORDS TO KNOW

product

the answer when you multiply

array

things set out in rows and columns

Division

THE BIG IDEA

Dividing means sharing into equal groups. It is the opposite of multiplying.

HOW TO DO IT

1. Read $12 \div 3$ as "12 shared into 3 equal groups".
2. Ask: what number times 3 makes 12?
3. Use your times tables to find it.
4. Check by multiplying your answer by the number you divided by.

WORKED EXAMPLES

$12 \div 3 = ?$

$3 \times 4 = 12$, so $12 \div 3 = 4$.

20 cookies are shared equally by 5 friends. How many each?

$5 \times 4 = 20$, so each friend gets 4 cookies.

WATCH OUT

Order matters. $12 \div 3$ is 4. The big number being shared goes first.

WORDS TO KNOW

quotient

the answer when you divide

fact family

$3 \times 4 = 12$, $4 \times 3 = 12$, $12 \div 3 = 4$, $12 \div 4 = 3$

Fractions

THE BIG IDEA

A fraction names equal parts of a whole. The bottom number says how many equal parts there are. The top number says how many parts you have.



3 of the 4 equal parts are shaded. That is $\frac{3}{4}$.

HOW TO DO IT

1. Count all the equal parts. That is the bottom number.
2. Count the shaded parts. That is the top number.
3. Same bottom numbers: the fraction with the bigger top is bigger.
4. Same top numbers: the fraction with the smaller bottom is bigger, because the pieces are bigger.

WORKED EXAMPLES

Which is bigger, $\frac{3}{8}$ or $\frac{5}{8}$?

The pieces are the same size. 5 pieces is more than 3. So $\frac{5}{8}$ is bigger.

Which is bigger, $\frac{1}{4}$ or $\frac{1}{8}$?

Cutting into 4 makes bigger pieces than cutting into 8. So $\frac{1}{4}$ is bigger.

WATCH OUT

The parts must be equal. A bigger bottom number means smaller pieces: $\frac{1}{8}$ is smaller than $\frac{1}{4}$.

WORDS TO KNOW

numerator

the top number

denominator

the bottom number

Area and perimeter

THE BIG IDEA

Perimeter is the distance around a shape. Area is the space inside it.



A rectangle 5 units long and 3 units wide.

HOW TO DO IT

1. For perimeter, add the lengths of all the sides.
2. For the area of a rectangle, multiply length $\times$ width.
3. Perimeter is measured in units. Area is measured in square units.
4. If you know the area and one side, divide to find the other side.

WORKED EXAMPLES

A rectangle is 5 by 3. Find the perimeter and the area.

Perimeter: $5 + 3 + 5 + 3 = 16$ units.

Area: $5 \times 3 = 15$ square units.

A rectangle has area 24. One side is 6. How long is the other side?

$24 \div 6 = 4$. The other side is 4.

WATCH OUT

Do not mix them up. Perimeter goes around, so you add. Area fills the inside, so you multiply.

WORDS TO KNOW

perimeter

the distance around a shape

area

the space inside a shape

Rounding

THE BIG IDEA

Rounding swaps a number for a nearby number that is easier to work with, like the nearest ten or hundred.

HOW TO DO IT

1. Find the place you are rounding to.
2. Look at the digit just to the right of it.
3. If that digit is 5 or more, round up. If it is 4 or less, stay the same.
4. Change all the digits to the right into zeros.

WORKED EXAMPLES

Round 47 to the nearest ten.

The ones digit is 7. That is 5 or more, so round up. The answer is 50.

Round 362 to the nearest hundred.

The tens digit is 6. That is 5 or more, so round up. The answer is 400.

WATCH OUT

5 rounds up. 45 rounds to 50, not 40.

WORDS TO KNOW

round

change to a nearby easier number

estimate

a close answer, not the exact one